

Multi-Omic Data Integration via Decentralized AI Agents for the Prediction of Phenotypic Drug Resistance in Cancer

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Introduction

Historically, the treatment of cancer was dictated primarily by anatomical origin, focusing on the question, “what type of cancer is this?”. However, the advent of precision oncology has fundamentally shifted this medical paradigm. Today, the critical clinical objective is no longer just identifying where the cancer originated, but rather answering, “what is unique about this particular cancer in this particular person?” Multi-omics, the integration of genomics, transcriptomics, proteomics, and epigenomics, plays an indispensable role in this shift. By mapping the intricate biological layers of a tumor, multi-omics provides a highly specific molecular fingerprint that dictates a patient's unique phenotype and potential for drug resistance.

To translate this massive, high-dimensional volume of multi-omic data into actionable treatment plans, the clinical gold standard is the Multidisciplinary Tumor Board (MDT). MDTs bring together diverse human specialists, such as oncologists, pathologists, geneticists, and pharmacologists, to debate, interpret, and synthesize complex, and often conflicting, clinical signals. Yet, as the dimensionality of biological data grows exponentially, human MDTs face severe constraints regarding scalability, throughput, and cognitive overload.

This study positions itself at the intersection of precision oncology and artificial intelligence by computationally modeling the MDT workflow. Currently, most AI applications in this space rely on monolithic Large Language Models (LLMs). However, when forced to synthesize massive, conflicting multi-omics payloads (e.g., genomics versus proteomics), single LLMs suffer from Context Overwhelm, frequently ignoring critical pathway bypass mechanisms and outputting highly confident but biologically inaccurate drug response predictions.

To resolve this, this research introduces a Decentralized Multi-Agent System (MAS) that acts as a council of cooperating AI specialists. By deploying independent agents that each evaluate distinct biological modalities, and requiring them to negotiate via a structured, consensus-driven Agent-to-Agent (A2A) protocol, this framework emulates the rigorous, peer-reviewed environment of a human tumor board. Ultimately, this approach aims to bridge the gap between complex multi-omic data and safe, auditable, and highly personalized clinical decision-making.

Literature Survey

MultiOmics in Precision Oncology is Now Well Established, and Why

The field of Cancer Drug Response (CDR) prediction has evolved from static chemical-structural analysis to sophisticated deep learning frameworks that treat drug-cell line interactions as a complex multimodal fusion problem (Saranya KR et al., 2024, Haochen Zhao et al., 2023, Chen, H.-O et al., 2024). Current research establishes a clear consensus that single-omics data is inherently limited due to the low correlation (0.40) between mRNA and protein levels and that integrating genomics, transcriptomics, and epigenomics significantly improves the refined accuracy of IC50 predictions (Du P et al., 2024, Saranya KR et al., 2024, Chen, H.-O et al., 2024, Liu XY et al., 2023). Landmark architectures like **DRN-CDR** and **MSDRP** have advanced the field by utilizing Deep Residual Networks (ResNets) and Similarity Network Fusion (SNF) to capture higher-order non-linear features and mitigate the curse of dimensionality (Saranya KR et al., 2024, Haochen Zhao et al., 2023, Almutiri et al., 2023, Liu XY et al., 2023). However, a pervasive “interpretability-performance trade-off” remains. While these black-box models achieve high AUC values, they frequently fail to provide the mechanistic reasoning required for clinical adoption, leaving the causal link between molecular features and therapeutic outcomes largely inferred (Haochen Zhao et al., 2023, Chen, H.-O et al., 2024, Jiang, W et al., 2025, Almutiri et al., 2023).

Pre-trained and Instruction Tuned LLMs for MultiOmics

Current research in generative AI for bioinformatics has transitioned from static feature engineering to powerful, task-specific architectures like AlphaFold and Enformer, which have revolutionized structural biology and genomic interpretation (Hein et al., 2025). However, these “black-box” models often struggle with multi-omic integration, frequently losing biological nuance when reconciling conflicting signals. While a microservices-driven paradigm (El-Sayed et al., 2026) improves computational scalability and throughput by up to **46%**, it still faces a semantic interoperability gap. Individual AI services can exchange data but cannot autonomously debate or prioritize signals based on the specific biological context of a tumor.

The transition from Automated Machine Learning (AutoML) to reasoning-based autonomy represents a pivotal shift in computational biology, as Large Language Model (LLM) agents begin to replicate the “plan-act-reflect” cycle of human experts (Martinek et al., 2025, Qu et al., 2025). Current research highlights that while multi-omics integration (combining genomics, transcriptomics, and proteomics) significantly outperforms single-omic models in predictive accuracy, it remains hindered by a profound Interpretability Gap (Zack M et al., 2025, Vavekanand et al., 2026). Landmark datasets like **Biology-Instructions** and platforms such as **PromptBio** have attempted to bridge this by instruction-tuning models on millions of biological sequences, yet these systems frequently encounter generalization collapse when faced with the 3D complexity of genomic data or the high-dimensional sparsity of single-cell omics (Martinek et al., 2025, Haonan He, et al., 2025, Ali S et al., 2025, Dip et al., 2025, Yang et al., 2025). Furthermore, a persistent biological bias exists due to a lack of population diversity in training

data, which leads to reduced generalizability across different ethnic groups (Zack M et al., 2025, Dip et al., 2025, Vavekanand et al., 2026).

LLMs as Treatment Recommenders

Recent scoping reviews and reproducibility studies (Zhou, S et al., 2025) (Meng X et al., 2024) (4) demonstrate that while LLMs show human-level potential in passing medical exams, their transition to real-world treatment recommendation is hindered by a clinical utility gap. Specifically, current models struggle with the iterative and messy nature of longitudinal Electronic Health Records (EHRs)(Zhou, S et al., 2025) (Williams C et al., 2024) and often exhibit an over-caution bias, as seen in large-scale studies where LLMs significantly lagged behind physicians in diagnostic specificity (Williams C et al., 2024) (Yang X et al., 2025). Furthermore, the hallucination and safety bottleneck remains a primary concern; even advanced models often favor probabilistic patterns over deep disease pathophysiology (Meng X et al., 2024) , leading to recommendations that may be linguistically fluent but biologically inaccurate. While hybrid neuro-symbolic architectures (Galitsky, B. A et al., 2024) and Retrieval-Augmented Generation (RAG)(Zhou, S et al., 2025)(Lin et al., 2025) have been proposed to mitigate these risks, most existing frameworks remain focused on clinical text summarization—reading doctor’s notes rather than calculating the underlying molecular drivers of a disease (Lin et al., 2025)(Williams C et al., 2024).

LLMs in Tumor Drug Response

The challenge of tumor drug resistance, which accounts for over 90% of cancer-related deaths, is fundamentally a problem of information lag and biological complexity that traditional statistical methods struggle to resolve (Mao Y et al., 2025). Current research has transitioned from single-modality analysis to multimodal AI models that integrate genomics, transcriptomics, and the tumor microenvironment to achieve high predictive accuracy (Mao Y et al., 2025, Liu et al., 2025, [X.Wu](#) et al., 2026, Yang Su et al., 2025). Landmark frameworks such as **CellHit** and **GeneRxGPT** have demonstrated that grounding Large Language Models (LLMs) in mechanistic pathway-aware features or real-time literature retrieval significantly outperforms static, black-box models (Carli, F. et al., 2025, Lai YJ et al., 2025). However, despite these advances, a significant translational gap remains. Models trained on *in vitro* cell lines often fail in real-world patients due to intra-tumor heterogeneity and the Black Box problem, where clinicians cannot decipher the underlying biological rationale behind a resistance prediction (Mao Y et al., 2025, Carli, F. et al., 2025, Yang Su et al., 2025).

LLM in MDT – single and multi-agent systems

The digital transformation of Multidisciplinary Tumor Boards (MDTs) has transitioned from monolithic, rule-based clinical decision support toward decentralized, multi-agent architectures that mirror the collaborative nature of human medical teams (Liu, Q. et al., 2026, Zhang et al., 2025, Han et al., 2025). Recent literature emphasizes that role-differentiated agents, specializing in diagnostics, treatment, and safety, significantly outperform single-model approaches by reducing context dilution and enhancing reasoning accountability (Liu, Q. et al., 2026, Zhang et al., 2025, Codella et al., 2025). Frameworks such as **EvoMDT** and **HAO** have demonstrated high factual recall and guideline concordance by utilizing specialized agents grounded in Retrieval-Augmented Generation (RAG) (Liu, Q. et al., 2026, Codella et al., 2025). However, despite these architectural gains, the field faces persistent bottlenecks of reasoning. Current benchmarks like **MTBBench** show that models perform near-randomly on complex longitudinal tasks like recurrence forecasting, and most systems struggle to move beyond surface-level text summarization to resolve the conflicting multi-omic signals inherent in precision oncology (Vasilev et al., 2025, Codella et al., 2025, Lorenzo et al., 2026).

Debate and Consensus Mechanisms in MAS

The integration of Large Language Models (LLMs) into bioinformatics has shifted the field from static algorithmic tools toward dynamic, reasoning-based multi-agent systems (MAS) capable of managing complex, multi-step workflows (Mehandru, N. et al., 2025, Yang T et al., 2025). As identified in recent frameworks like **BioAgents**, while single-agent systems excel at conceptual genomics and tool selection, they frequently struggle with the functional execution of code and the procedural knowledge gap required for end-to-end analysis (Mehandru, N. et al., 2025). Furthermore, the iterative refinement process often touted in multi-agent collaboration has shown diminishing returns. Repeated self-correction rounds can lead to knowledge inertia or even degrade output quality, suggesting that linguistic debate is not a substitute for grounding in high-quality domain data (Mehandru, N. et al., 2025, Zhou J et al., 2025).

This limitation is closely tied to the architectural tension between collaborative consensus and voting paradigms. While traditional Multi-Agent Debate (MAD) frameworks treat these protocols as static variables, recent empirical evaluations demonstrate a sharp performance dichotomy: consensus protocols show a performance improvement of 2.8% on factual, knowledge-dense tasks, whereas voting protocols yield a 13.2% optimization on abstract reasoning tasks (Kaesberg et al., 2025). However, scaling conversational rounds blindly deteriorates system accuracy due to “problem drift” and “sycophantic convergence” (Kaesberg et al., 2025), alongside an inherent unreliability tax where multi-agent frameworks consume 15 to 50 times more tokens than standalone models (Spieser J et al., 2026). To counteract this computational overhead, contemporary architectures utilize heterogeneous agent groups to introduce epistemic diversity and combat shared bias (Liu, Y et al., 2026). They also deploy sparse, progressive communication topologies such as localized intra-group debate paired with hierarchical inter-group synthesis which can reduce total token expenditure by up to 46.9% without sacrificing logical reasoning depth (Liu, T. et al., 2024).

Crucially, standard consensus-driven aggregation models rely heavily on a final “judge” model or majority voting, working under the flawed assumption that agreement equates to biological accuracy (Cui, Y. et al., 2025). This architecture is highly susceptible to the “Silent Agreement” problem, where systemic conformity and herding effects suppress critical, independent reasoning paths (Cui, Y. et al., 2025). When no clear consensus exists, standard majority voting becomes statistically random and inequitable (Cui, Y. et al., 2025). While advanced frameworks have introduced mathematically rigorous aggregation tools, such as using Kendall’s coefficient of concordance (W) to identify discordant agents or applying multi-agent reinforcement learning (PPO) to optimize consensus paths, these methods frequently experience latency spikes and require up to 252 GB of memory (Han, X. et al., 2025).

This technical barrier separates the current lab-to-live execution gap, where systems performing at a 90% accuracy tier on static examinations experience a catastrophic drop to 45 - 69% when subjected to practice-based, unstructured clinical data (Spieser J et al., 2026). Furthermore, existing autonomous workflows remain largely reactive and restricted by a top-down supervisor-worker topology, such as PromptBio’s supervisor routing engine, which exhibits severe performance degradation when generating code for complex, multi-step survival analyses (Yang et al., 2025). This underscores the critical need for decentralized, peer-to-peer agent interaction spaces that implement native, reasoning-heavy backbones alongside standardized interfaces to safely bridge data operations with true biological causality.

MCPs in Bioinformatics and MAD

The transition from human-mediated command-line tools to autonomous, agent-ready infrastructures represents a critical frontier in computational biology (Widjaja et al., 2025, Flotho M et al., 2026). As identified in recent frameworks like **BioinfoMCP**, the bottleneck in AI-driven science has shifted from raw reasoning capability to the manual engineering required to interface Large Language Models (LLMs) with legacy bioinformatics tools (Widjaja et al., 2025). While the Model Context Protocol (MCP) provides a standardized semantic contract that moves beyond basic APIs to enable machine-actionable data stewardship, current implementations often struggle with Hardware-Software Co-optimization and a lack of robust Error-Aware Recovery for biological edge cases (Widjaja et al., 2025, Flotho M et al., 2026). Furthermore, a significant research gap remains in providing an audit trail for high-stakes clinical scenarios, as most current MCP-enabled agents focus on functional execution rather than the mechanistic explainability required for regulatory trust (Flotho M et al., 2026).

The shift from single-model prompting toward collaborative multi-agent architectures has highlighted the potential of **Multi-Agent Debate (MAD)** to mitigate hallucinations and enhance reasoning through iterative peer review. However, as noted in the **M3MAD-Bench** study (li et al., 2026), current frameworks often suffer from divergent noise in unimodal tasks and a plateau effect, where simply increasing the rounds of debate fails to reach a more accurate convergence. A critical research gap remains in the reliance on fixed roles and rigid protocols that fail to capture **environment-driven feedback**, leading to “collective delusion” where agents

mutually reinforce incorrect assumptions due to a lack of grounding in external, high-fidelity data.

Research Contribution

The preceding review identifies a critical divergence in the field of precision oncology. While predictive accuracy in multi-omic integration is rising, clinical interpretability and causal grounding remain stagnant. Current architectures, ranging from monolithic LLMs to RAG-based clinical assistants, consistently struggle with probabilistic hallucination, context dilution, and a reliance on surface-level text rather than raw biological data. To bridge this gap, this thesis introduces a Decentralized Multi-Agent Consensus Framework. By utilizing the Model Context Protocol (MCP), the framework creates a live semantic bridge to structured databases (e.g., GDSC, CCLE, and cBioPortal), enabling agents to bypass uncurated text and reason directly over raw numerical biomarkers like Z-scores and mutation frequencies.

The core innovation of this work lies in the transition from open-ended linguistic debate to a **Hierarchy of Truth** protocol. Unlike traditional multi-agent systems that rely on weighted voting or LLM-as-a-judge mechanisms, this framework implements a deterministic consensus logic grounded in the Central Dogma of molecular biology. By requiring agents to prove mechanistic overrides, such as a DNA structural binding failure physically nullifying a transcriptomic sensitivity signal, the system replicates the deliberative rigor of a Multidisciplinary Tumor Board (MDT). This approach directly solves the “black-box” problem and the “cold-start” challenge for new drugs by grounding every clinical verdict in verifiable biological axioms rather than pattern-matching. In doing so, this research provides a scalable, auditable, and biologically faithful architecture for the next generation of personalized cancer therapy.

Research Gap Addressed

1. The "Context Overwhelm" Problem

Current AI applications in drug resistance generally fall into two categories, both of which have significant flaws:

- **Monolithic LLMs:** Most researchers use a single, large model (like GPT-4) to predict outcomes. However, when you feed one model DNA, RNA, and Pathway data simultaneously, the model suffers from **Attention Bias**. It often "latches on" to a well-known DNA mutation (like BRAF) and ignores the subtle RNA "bypass" signals, leading to reasoning hallucinations.

- **Sequential Pipelines:** Many tools use linear pipelines (Agent A → Agent B). If Agent A makes a mistake in gene extraction, that error is passed down the chain without any cross-examination, leading to an overall incorrect result.
- **Pre-trained and Instruction-tuned LLMs:** To power the specialized agents within the decentralized council, this framework leverages a hybrid suite of pre-trained and instruction-tuned Large Language Models (LLMs) tailored to specific cognitive tasks. For instance, GPT-4o is utilized for high-level clinical reasoning, debate synthesis, and consensus evaluation due to its superior context retention and logical alignment. Concurrently, models such as Claude 3.5 Sonnet are deployed for deterministic routing, JSON parsing, and rapid tool execution (A2T operations), optimizing both computational efficiency and reasoning accuracy across the multi-agent system.

2. The Novelty: What Makes Our Approach Different?

The new architecture introduces three specific innovations that do not exist in the current literature:

A. Decentralized "Council" vs. Centralized Brain

Instead of one AI trying to be a genius in everything, we isolate expertise by domain. So, the Genomics Agent *only* sees DNA, and the Pathways Agent *only* sees the network map. They are forced to reconcile their different "truths" through a structured debate, mimicking a real-world multi-disciplinary tumor board.

B. Agent-to-Tool (A2T) Protocol (via MCP)

While Large Language Models possess broad generalized knowledge, precision oncology requires high-fidelity, up-to-date molecular data. To bridge this gap, the system utilizes the Model Context Protocol (MCP) as its Agent-to-Tool (A2T) communication layer. It is important to note that MCP serves as the established mechanism (the underlying infrastructural pattern) for executing API calls and structured database queries. By providing a secure, standardized handshake with external repositories like the cBioPortal (for genomics) and the BioThings CCLE API (for expression), the A2T protocol ensures the agents are dynamically fetching raw molecular values (e.g., Z-scores and IC50). This guarantees that the subsequent inter-agent reasoning is grounded in empirical, live data rather than static, scraped literature.

C. Plug-and-Play System

Since every individual agent is learning independently and eventually becomes an expert in the database that they are learning from, it is fairly simple to add another agent to enhance the Explainable output we need. However, we do need to consider the impact on compute resources and the time it takes to generate a response on adding multiple different agents to handle individual tasks.

D. Agent-to-Agent (A2A) Consensus Protocol

The core architectural novelty of this research lies in the inter-agent communication layer, or the Agent-to-Agent (A2A) protocol. Rather than relying on a simple critic-reviewer loop or basic majority voting, the A2A protocol orchestrates advanced debate, strategic planning, and rigorous consensus formation among the sovereign agents. When conflicting multi-omic signals arise (such as a genomic mutation suggesting drug sensitivity while a pathway map indicates a bypass mechanism), the agents communicate via a structured JSON message bus. They negotiate these biological contradictions using a deterministic “Hierarchy of Truth” (e.g., prioritizing structural binding failures over transcriptomic noise). This A2A collaboration explicitly models the deliberative workflow of a human Multidisciplinary Tumor Board (MDT), ensuring that the final clinical recommendation is the result of a biologically validated consensus rather than LLM sycophancy or probabilistic hallucination.

For example,

User query: “Evaluate the predicted response of Vemurafenib for the COLO205 cell line. We know it has a BRAF V600E mutation, but we want to make sure it won't resist.”

Genomic Agent: Confirms that the BRAF V600E mutation is present. It reports that the target is “**sensitive**” to Vemurafenib.

Expression Agent: Discovers that the **EGFR gene** is significantly overexpressed (high z-score) in this specific cell line (COLO25). It reports “**neutral**” as it sees the target, but it also sees an unusually 'loud' EGFR signal.

Pharmacology Agent: Pulls the historical **LN_IC50** for Vemurafenib. It reports “**sensitive**” as historically, COLO205 shows moderate sensitivity in baseline lab tests.

Pathways Agent: Maps the network and sees that **EGFR** sits “downstream” or parallel to BRAF, potentially feeding the same growth signal. It reports “**resistant**” because, if EGFR is active, it won't affect the signal whether BRAF is mutated or not and the drug won't work.

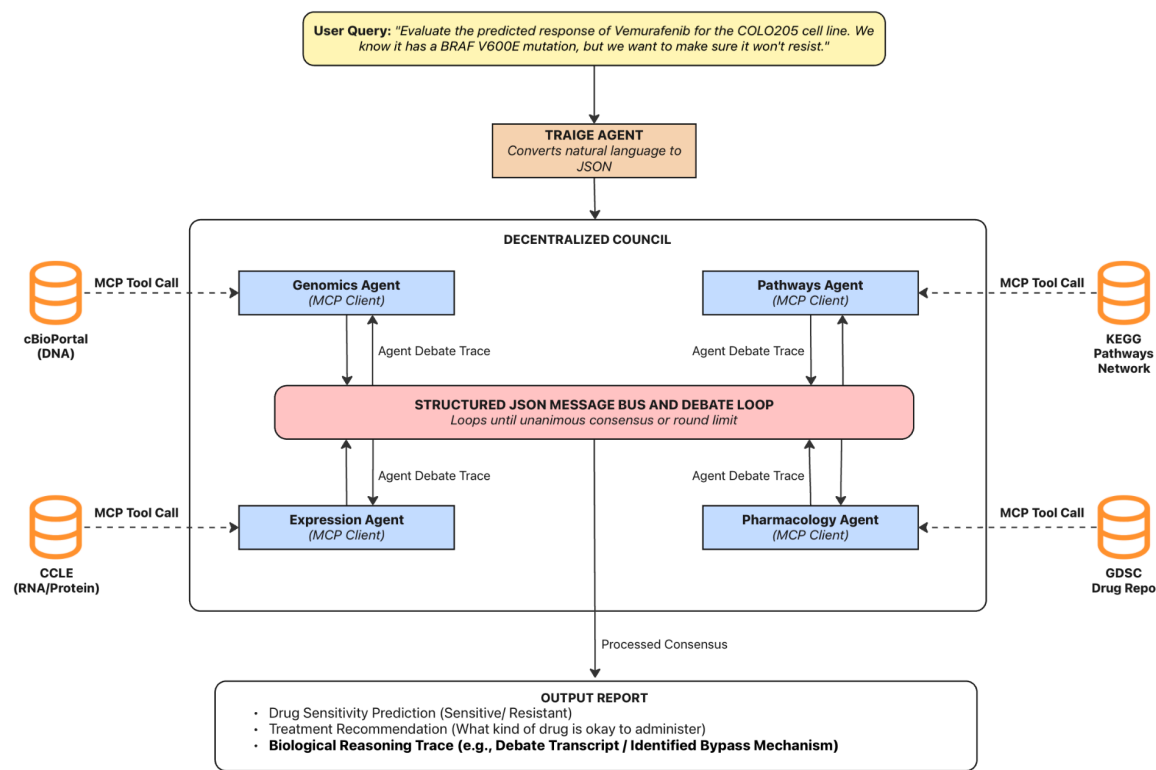
Now to finally decide the verdict, we generally just add up the “**sensitive**”, “**resistant**”, and “**neutral**” scores together. So, ideally the LLM responds back that the target is sensitive (as it has the highest score). But this is where we see the hallucination. The LLM has skipped past the message from the Pathways agent that there is an alternative path that keeps the signal active even if BRAF is mutated. This is the flaw we wish to address by enabling a debate among the agents where they see each other's responses and arrive at a consensus to make the final decision.

Our Solution

This study proposes a novel **Decentralized Multi-Agent Council** architecture. Rather than relying on a single, overwhelmed LLM, the system isolates biological modalities into distinct,

specialized AI agents. Using the Model Context Protocol (MCP), each agent is hardwired strictly to its domain-specific, live database.

To eliminate the "Sycophancy" (people-pleasing) flaw of hierarchical Judge models, the agents operate as a peer-to-peer council. They utilize a structured JSON-based communication protocol to debate conflicting data over a shared asynchronous message bus until a biological consensus is reached.



Brief Methodology

This research implements a decentralized, multi-agent AI architecture for precision oncology, utilizing the Model Context Protocol (MCP) to interface with four distinct biological data modalities. The study transitions from traditional text-based retrieval to a federated API approach, where reasoning is grounded in live, structured datasets.

Phase 1: Input Orchestration (The Triage Agent)

The system employs a **Triage Agent** as a pre-processor to convert natural language queries into a machine-readable JSON input.

- **Entity Extraction:** The agent extracts the `subject_id` (e.g., COLO205) and `target_drug` (e.g., Vemurafenib).
- **Route Determination:** Based on these parameters, the Triage Agent selects the appropriate API endpoints for the subsequent Council debate, mapping common cell line names to standardized **Sanger Model IDs** or **COSMIC IDs**.

Phase 2: Agent-Specific Data Acquisition (The API Layer)

Four specialized agents operate as independent MCP clients, querying the following first-party and production-grade APIs:

1. **Genomics Agent:**
 - **Interface:** cBioPortal MCP Server.
 - **API Target:** <https://www.cbioportal.org/api>.
 - **Primary Tool:** `get_mutations` for fetching non-synonymous DNA variants and `get_copy_number_alterations` to identify oncogenic amplifications (e.g., EGFR).
2. **Expression Agent:**
 - **Interface:** BioThings CCLE API.
 - **API Target:** <https://biothings.ncats.io/ccle>.
 - **Primary Tool:** `/query` endpoint to retrieve z-scored RNA-Seq and Proteomic expression data for the specific `subject_id`.
3. **Pathways Agent :**
 - **Interface:** KEGG REST API.
 - **API Target:** http://rest.kegg.jp/get/{pathway_id}/kgml.
 - **Primary Tool:** Fetches **KGML (XML)** network files. A local Python-based parser (using `xml.etree`) converts the XML into a directional graph (edges/nodes) for LLM consumption, identifying downstream bypass mechanisms.
4. **Pharmacology Agent:**
 - **Interface:** Sanger DepMap / Cell Model Passports API.
 - **API Target:** <https://api.cellmodelpassports.sanger.ac.uk>.
 - **Primary Tool:** `/models/{id}/datasets/gdsc` to retrieve historical **LN_IC50** values and drug sensitivity highlights.

Phase 3: The Agent-to-Agent (A2A) Debate Protocol

The agents enter a multi-round deliberation cycle via a **Structured JSON Message Bus**.

- **Domain Isolation:** Each agent only "sees" the API data from its specific domain until the debate begins.
- **Logical Synthesis:** Agents must justify their prediction (Sensitive or Resistant) by citing specific data points retrieved from their respective APIs.

Phase 4: Evaluation and Validation

- **Quantitative Benchmark:** System accuracy is measured against a hold-out set of known GDSC outcomes using Precision and Recall.
- **Qualitative Analysis:** The "Agent Debate Trace" is evaluated for biological accuracy, ensuring the reasoning follows established molecular biology signaling cascades (e.g., verifying that a BRAF block is correctly identified as failing due to an EGFR bypass).

This methodology establishes a rigorous framework for an **Explainable AI (XAI)** system that moves beyond simple text generation into factual, data-driven oncological reasoning.

Validation

To ensure the integrity of the validation process and avoid data leakage, the Pharmacology Agent will utilize the GDSC2 dataset for historical IC50 baselining, while the system's final predictive accuracy will be benchmarked against independent external datasets, specifically CTRPv2 and curated clinical case studies in the form of published research articles.

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